

Build a professional AI-powered education platform called **LAMBERTIQ**.

BRAND

Name: LAMBERTIQ

Tagline: Learn Smarter. Think Better. Achieve More.

LAMBERTIQ is a multi-model AI study platform designed to help students learn, revise, practice and understand academic subjects.

The platform must support both **English and Kiswahili**.

1. MULTI-AI ENGINE

Do NOT build LAMBERTIQ around only one AI provider.

Create a modular **Multi-AI Provider System**.

Architecture:

Student



LAMBERTIQ AI Router



AI Provider Manager

├— Google Gemini

├— Groq

├— OpenRouter

├— Mistral

├— Hugging Face

├— Cloudflare Workers AI

└— Other compatible providers



Response Collector



Response Analyzer



AI Synthesizer



Final Answer

The system must allow administrators to enable/disable providers.

The architecture must make it easy to add new AI providers later.

2. FREE AI API SUPPORT

Design the application to support free-tier APIs wherever available.

Initial provider adapters:

Google Gemini

Environment variable:

GEMINI_API_KEY

Groq

Environment variable:

GROQ_API_KEY

OpenRouter

Environment variable:

OPENROUTER_API_KEY

Mistral

Environment variable:

MISTRAL_API_KEY

Hugging Face

Environment variable:

HUGGINGFACE_API_KEY

Cloudflare Workers AI

Environment variables:

CLOUDFLARE_ACCOUNT_ID

CLOUDFLARE_API_TOKEN

Do not expose API keys in the frontend.

Use server-side API routes/functions and environment variables.

IMPORTANT:

Never hard-code API keys.

Free-tier limits must be configurable because provider quotas and available models can change.

3. AI ROUTER

Create an intelligent AI Router.

The router should determine which models should answer a question.

Example:

Simple question:

→ 1 AI model

Complex academic question:

→ 2–4 AI models

Engineering calculation:

→ reasoning-capable models + calculator/validation layer

Document-based question:

→ models with appropriate context length

Important academic question:

→ multiple models + synthesis

Do not always call every provider because this would unnecessarily consume free quotas.

4. MULTI-MODEL COMPARISON

When multiple models answer a question:

Collect:

- Model name
- Provider
- Answer
- Response time
- Token usage if available

- Status
- Errors
- Confidence metadata where available

Then compare the responses.

The synthesizer should:

1. Identify agreements.
2. Identify contradictions.
3. Identify missing information.
4. Check calculations where possible.
5. Prefer information supported by the student's uploaded material.
6. Avoid inventing information.
7. Clearly indicate uncertainty.
8. Produce one understandable final answer.

Do NOT assume that the majority answer is automatically correct.

5. ENVIRONMENTAL ENGINEERING

Create a dedicated **Environmental Engineering** department.

Subjects should include:

Core Environmental Engineering

- Principles of Environmental Engineering
- Environmental Chemistry
- Environmental Biology
- Environmental Microbiology
- Environmental Engineering Mathematics
- Environmental Engineering Hydraulics
- Fluid Mechanics

- Environmental Engineering Laboratory
- Environmental Engineering Design
- Environmental Health
- Environmental Risk Assessment

Water

- Water Supply Engineering
- Drinking Water Treatment
- Water Quality
- Water Treatment Plant Design
- Water Distribution Systems
- Hydrology
- Surface Water Pollution
- Groundwater
- Water Quality Modelling

Wastewater

- Wastewater Engineering
- Wastewater Collection Systems
- Domestic Wastewater Treatment
- Industrial Wastewater Treatment
- Biological Treatment
- Physical-Chemical Treatment
- Sludge Treatment
- Sludge Disposal
- Wastewater Treatment Plant Design

Air Pollution

- Air Pollution

- Air Quality
- Atmospheric Pollution
- Emission Sources
- Air Pollution Control
- Atmospheric Dispersion
- Indoor Air Quality
- Air Quality Monitoring

Solid & Hazardous Waste

- Solid Waste Management
- Integrated Solid Waste Management
- Municipal Waste
- Industrial Waste
- Hazardous Waste
- Medical Waste
- Recycling
- Composting
- Landfill Design
- Waste-to-Energy

Soil & Groundwater

- Soil Pollution
- Groundwater Pollution
- Contaminant Transport
- Site Remediation
- Soil and Groundwater Monitoring
- Environmental Geology

Environmental Management

- Environmental Impact Assessment (EIA)
- Environmental Auditing
- Environmental Management Systems
- ISO 14001
- Environmental Policy
- Environmental Law
- Occupational Health and Safety
- Environmental Risk Management
- Climate Change
- Sustainability
- Environmental Management

Engineering Mathematics & Science

- Calculus
- Differential Equations
- Statistics
- Probability
- Physics
- Chemistry
- Biology
- Numerical Methods
- Engineering Mathematics

6. ENVIRONMENTAL ENGINEERING AI TUTOR

Create a specialized tutor mode:

Environmental Engineering Tutor

Students should be able to ask:

"Explain BOD and COD."

"Design a simple wastewater treatment process."

"Explain activated sludge."

"What is coagulation and flocculation?"

"Calculate hydraulic retention time."

"Explain Environmental Impact Assessment."

"Give me an example of a solid waste management plan."

"Explain air pollution control technologies."

"Help me understand groundwater contamination."

The AI should explain engineering concepts step-by-step.

7. ENGINEERING CALCULATION MODE

Create a special:

Engineering Solver

It should help students with:

- Equations
- Unit conversions
- Mass balance
- Energy balance
- Flow calculations
- Hydraulic calculations
- Water treatment calculations
- Wastewater calculations
- Environmental chemistry calculations
- Statistics
- Engineering mathematics

The system must show:

Given Data

↓

Formula

↓

Substitution

↓

Calculation

↓

Units

↓

Final Answer

↓

Explanation

Do not simply provide the final number.

For numerical calculations, use a deterministic calculation/validation layer where possible instead of relying entirely on an LLM.

8. DOCUMENT ANALYZER

Students can upload:

- PDF
- DOCX
- PPTX
- TXT

Example:

Student uploads:

"Environmental Engineering Lecture Notes.pdf"

Then asks:

"Explain activated sludge according to my notes."

The system should prioritize the uploaded document.

Response should distinguish:

According to your document

and

Additional explanation

9. AI STUDY MODES

Create:

Ask Tutor

Normal AI conversation.

Teach Me

AI teaches the topic step-by-step.

Explain Simply

Converts complex engineering concepts into beginner-friendly explanations.

Exam Mode

Prepares students for examinations.

Quiz Mode

Generates questions and evaluates answers.

Calculation Mode

Solves engineering calculations step-by-step.

Revision Mode

Creates revision notes.

Document Q&A

Answers questions from uploaded materials.

Research Mode

Helps students explore academic topics while clearly distinguishing verified sources from AI-generated explanation.

10. QUIZ GENERATOR

Generate:

- Multiple Choice Questions
- True/False
- Short Answer
- Calculation Questions
- Engineering Problems
- Exam-style Questions

Difficulty:

Easy

Medium

Hard

University Level

Allow selection of:

Subject

Topic

Number of questions

Difficulty

Language

11. ENVIRONMENTAL ENGINEERING QUESTION BANK

Create a question bank organized by:

Department



Subject



Topic



Difficulty

Example:

Environmental Engineering
→ Wastewater Engineering
→ Activated Sludge
→ Medium
→ Question

Administrators can add/edit/delete questions.

12. STUDY DASHBOARD

Create:

Dashboard
AI Tutor
My Documents
Subjects
Environmental Engineering
Quizzes
Question Bank
Study Planner
Progress
Settings

Dashboard should show:

- Questions asked
 - Topics studied
 - Quiz scores
 - Study streak
 - Recent documents
 - Recommended topics
 - Weak topics
 - Study time
-

13. SMART STUDY RECOMMENDATION

Analyze student activity.

If a student performs poorly in:

"Water Treatment"

recommend:

- Water Quality
- Coagulation
- Flocculation
- Sedimentation
- Filtration
- Disinfection

Create personalized study recommendations.

14. ADMIN DASHBOARD

Admin should manage:

- Students
- Subjects
- Departments
- Topics
- AI Providers
- AI Models
- API usage
- AI routing
- Documents
- Quiz questions
- Study materials

- System prompts
- Usage limits
- Analytics

15. AI PROVIDER MANAGEMENT

Create an admin table:

Provider

Model

Status

Priority

Free/Paid

Daily Limit

Requests Used

Last Error

Example:

Gemini | Flash model | Active

Groq | Open model | Active

OpenRouter | Free model | Active

Mistral | Free model | Active

Hugging Face | Open model | Active

Cloudflare | Open model | Active

The system should NOT assume these models or quotas remain unchanged.

Make provider/model configuration dynamic.

16. FALLBACK SYSTEM

If:

Gemini fails

↓

Try Groq

If Groq fails

↓

Try OpenRouter

If OpenRouter fails

↓

Try Mistral/Hugging Face

If all fail:

Return a friendly error.

Never expose raw API keys, stack traces or sensitive provider information to students.

17. COST CONTROL

Because this project should initially operate with free AI APIs:

Implement:

- Daily user limits
- Per-minute limits
- Token limits
- Provider quotas
- Model priority
- Automatic fallback
- Response caching
- Request deduplication
- Admin usage monitoring

Allow administrators to define:

FREE USER LIMIT

PREMIUM USER LIMIT

ADMIN LIMIT

18. AI RESPONSE UI

Show:

LAMBERTIQ AI

Final synthesized answer.

Optional:

AI Models Consulted

✓ Gemini

✓ Groq

✓ OpenRouter

Then allow:

View Model Responses

Students can compare individual responses if desired.

Do not clutter the main answer with raw AI outputs.

19. TRUST & ACADEMIC SAFETY

For engineering subjects:

The AI should clearly state when an answer needs verification.

For calculations:

Show formulas and units.

For academic research:

Encourage students to verify important claims using textbooks, lecturers and authoritative sources.

Do not fabricate references.

Do not create fake citations.

20. DATABASE

Create tables:

users
departments
subjects
topics
conversations
messages
documents
document_chunks
ai_providers
ai_models
ai_requests
ai_responses
synthesized_answers
quizzes
quiz_questions
quiz_attempts
study_progress
study_plans
question_bank
system_settings

Use proper indexes and relationships.

21. INITIAL DEPARTMENTS

Start with:

1. Environmental Engineering
2. Computer Science
3. Information Technology
4. Software Engineering
5. Civil Engineering
6. Electrical Engineering
7. Mechanical Engineering
8. Business

- 9. Accounting
- 10. Economics
- 11. Mathematics
- 12. Statistics

Allow administrators to add more departments.

22. LANGUAGE

Support:

English
Kiswahili

Examples:

"Explain wastewater treatment in Kiswahili."

"Explain BOD and COD using a simple example."

"Teach me Environmental Impact Assessment in English."

The AI should preserve technical terminology while explaining difficult concepts in the selected language.

23. UI/UX

Create a premium modern education platform.

Style:

- Clean
- Professional
- Modern
- Minimal
- Mobile-first
- Responsive
- Student-friendly

Main branding:

LAMBERTIQ

Use a professional AI/education visual identity.

24. SECURITY

Never expose:

- API keys
- Provider secrets
- Database credentials
- Admin credentials

Use:

Environment variables

Server-side API routes

Authentication

Authorization

Input validation

Rate limiting

25. IMPORTANT ARCHITECTURE

Use this structure:

Frontend

↓

Backend/API

↓

AI Router

↓

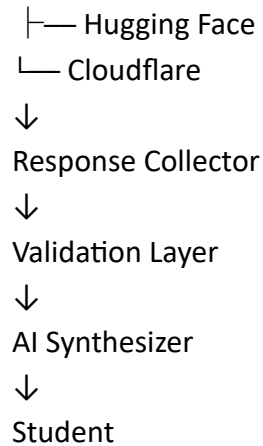
Provider Adapters

├— Gemini

├— Groq

├— OpenRouter

├— Mistral



This architecture must make it possible to add another AI provider later without rewriting the application.

FINAL PRODUCT

The finished product should be a real **Multi-AI Academic Learning Platform**, not simply a chatbot.

The major differentiator is:

LAMBERTIQ brings multiple AI models together to create a personalized learning environment for students, with a strong focus on engineering education and Environmental Engineering.

Start development in phases:

PHASE 1 — UI and project architecture

PHASE 2 — Database

PHASE 3 — Authentication

PHASE 4 — AI Provider System

PHASE 5 — Multi-AI Router

PHASE 6 — AI Synthesis

PHASE 7 — Environmental Engineering subjects

PHASE 8 — Document Analyzer

PHASE 9 — Quiz & Question Bank

PHASE 10 — Engineering Calculation Solver

PHASE 11 — Student Dashboard

PHASE 12 — Admin Dashboard

PHASE 13 — Testing

PHASE 14 — Deployment

Do not skip architectural foundations.